

Introduction to Artificial Intelligence

Assignment 1: AI System Specification

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Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Selected System: C — Agricultural Monitoring Drone for Crop Disease Detection

Task 1: PEAS Specification

The selected AI system is an Agricultural Monitoring Drone designed to fly over farmland and automatically detect early signs of crop diseases. Below is the complete PEAS breakdown:

PEAS Framework — Agricultural Monitoring Drone	
P — Performance Measure	- Disease Detection Accuracy: Percentage of infected crop patches correctly identified (target $\geq 90\%$) - False Alarm Rate: Number of healthy plants incorrectly flagged as diseased (target $< 5\%$) - Coverage per Flight Hour: Area of farmland scanned per hour (target ≥ 50 acres/hour) - Battery Efficiency: Flight time maintained per charge cycle (target ≥ 45 min) - Early Detection Rate: Diseases caught at Stage 1 or 2 before spreading to neighboring plants
E — Environment	Large open agricultural fields, variable weather conditions (wind, humidity, sunlight), uneven terrain, and different crop types (wheat, rice, cotton). The drone also operates in both morning and evening lighting conditions.
A — Actuators	Flight controller motors (for navigation and altitude control), high-resolution RGB camera gimbal (for stable image capture), infrared/multispectral camera (for detecting plant stress not visible to the naked eye), GPS-guided waypoint system (for systematic field coverage), and a wireless data transmitter (to send alerts to farmers in real time).
S — Sensors	RGB camera (24 MP, for visible disease symptoms like spots and discoloration), infrared camera (for detecting heat stress and water deficiency), GPS module (for precise location tagging of infected zones), altimeter and barometer (for maintaining stable flight height), wind speed sensor (to adjust flight path in windy conditions), and onboard temperature sensor.

Task 2: Environment Classification

Below is the classification of the task environment along six standard dimensions, with justification for each choice:

Dimension	Classification	One-Sentence Justification
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Observability	Partially Observable	The drone's cameras cannot see underground root conditions or disease hidden beneath dense leaf canopies.
Agent Type	Single-Agent	Only one primary decision-making drone is active at a time; multiple drones would form a multi-agent system.
Determinism	Stochastic	Wind gusts, cloud cover, and sudden lighting changes make outcomes unpredictable even for the same action.
Episode Type	Sequential	Each flight decision (where to scan next) depends on previously scanned regions and detected disease patterns.
Dynamism	Dynamic	The environment changes during operation — clouds shift, wind speeds vary, and crop conditions evolve in real time.
Continuity	Continuous	The drone deals with continuous sensor readings (GPS coordinates, image pixels, altitude) rather than fixed discrete states.

Task 3: Critical Analysis

Question 1: Greatest Technical Challenge

Among the six dimensions, the Stochastic (non-deterministic) nature of the environment poses the greatest technical challenge. Unlike a controlled lab environment, the agricultural field is constantly changing. Wind speeds affect the drone's camera stability, making images blurry. Sudden changes in sunlight create false shadows on crops, which the AI might incorrectly classify as diseased patches. Humidity changes can also interfere with infrared sensor readings. These unpredictable variables mean the AI system can never rely on fixed input-output rules — it must use probabilistic models (like Bayesian classifiers or trained CNNs) that handle uncertainty gracefully. This makes training, testing, and deployment significantly harder compared to a deterministic environment.

Question 2: Utility Function and Trade-off (Speed vs. Accuracy)

A suitable utility function for this drone agent could be defined as:

$$U = 0.6 \times \text{Detection_Accuracy} - 0.4 \times \text{Energy_Cost}$$

This function gives more weight to accuracy (0.6) than to energy saving (0.4), reflecting the real-world priority — missing a disease outbreak is more costly than using extra battery power. If the weight of accuracy were doubled (e.g., $0.8 \times \text{Accuracy} - 0.2 \times \text{Energy}$), the drone's behavior would change significantly: it would fly more slowly to capture clearer images, take multiple passes over suspicious regions, and choose slower but more precise scanning paths. This would improve detection rates but reduce the total area covered per flight, meaning the farmer would need more flights to cover the full field. The agent would essentially trade off speed for quality, which makes sense during high-risk disease seasons but would be inefficient for routine weekly monitoring.

Task 4: Structured Data Design (JSON Practice)

Below is the valid JSON representation of the system defined in Tasks 1 and 2:

```
{
  "system_name": "Agricultural Monitoring Drone",
  "peas": {
    "performance_measures": [
      "disease_detection_accuracy",
      "false_alarm_rate",
      "coverage_per_flight_hour",
      "battery_efficiency",
      "early_detection_rate"
    ],
    "environment": "Large open farmland with variable weather, wind, and lighting",
    "actuators": [
      "flight_controller",
      "RGB_camera_gimbal",
      "infrared_camera",
      "GPS_waypoint_system",
      "wireless_transmitter"
    ],
    "sensors": [
      "RGB_camera",
      "infrared_camera",
      "GPS_module",
      "altimeter",
      "wind_speed_sensor",
      "temperature_sensor"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Hidden root and canopy conditions are not visible to the drone cameras"
    },
    "agents": {
      "choice": "Single-agent",
      "justification": "One primary decision-making drone operates at a time"
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "Wind and lighting changes make outcomes unpredictable"
    },
    "episode_type": {
      "choice": "Sequential",
      "justification": "Each scanning decision depends on previously covered regions"
    },
    "dynamism": {
      "choice": "Dynamic",
```

```
        "justification": "Clouds, wind, and crop conditions change while the drone  
is flying"  
    },  
    "continuity": {  
        "choice": "Continuous",  
        "justification": "Sensor readings are real-valued and continuous, not  
discrete"  
    }  
},  
"utility_function": "U = 0.6 * detection_accuracy - 0.4 * energy_cost"  
}
```